

ELITE IGCSE ACADEMY

Student Past Paper Solutions

Jan 2022 P1HR Worked Solutions

Jan 2022 | P1HR

29 questions ■ question image followed by worked solution

PREPARED BY

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STUDENT EDITION

Worked answers included

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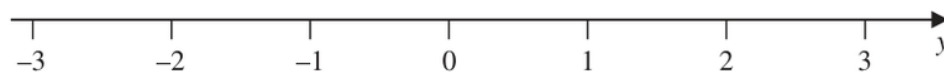
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PAST PAPER QUESTION**Q#: 1** **Marks: 3****TOPIC: Solving Inequalities**

Jan 2022 P1HR - Jan 2022 | P1HR

1 n is an integer.(a) Write down all the values of n such that $-2 \leq n < 3$

(2)

(b) On the number line, represent the inequality $y \leq 1$ 

(1)

(Total for Question 1 is 3 marks)

EA

PAST PAPER SOLUTION**Q#: 1****Marks: 3****TOPIC: Solving Inequalities**

Jan 2022 P1HR - Jan 2022 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. (a)**

(a)

$$-2 \leq n < 3$$

2. Calculate valueSince n is an integer,

$$n = -2, -1, 0, 1, 2$$

3. Solve inequality

(b) For

$$y \leq 1$$

4. use a closed circle at

use a closed circle at 1 and shade to the left.

FINAL ANSWER $n = -2, -1, 0, 1, 2$; closed at 1, shaded left.

PAST PAPER QUESTION**Q#: 2****Marks: 2****TOPIC: Probability Toolkit**

Jan 2022 P1HR - Jan 2022 | P1HR

2 Each time John plays a game, the probability that he wins the game is 0.65

John is going to play the game 300 times.

Work out an estimate for the number of games that John wins.

(Total for Question 2 is 2 marks)

EA

PAST PAPER SOLUTION

Q#: 2

Marks: 2

TOPIC: **Probability Toolkit**

Jan 2022 P1HR - Jan 2022 | P1HR

WORKED SOLUTION

Dr. Eslam Ahmed

1. Calculate probability

$$0.65 \times 300 = 195$$

FINAL ANSWER

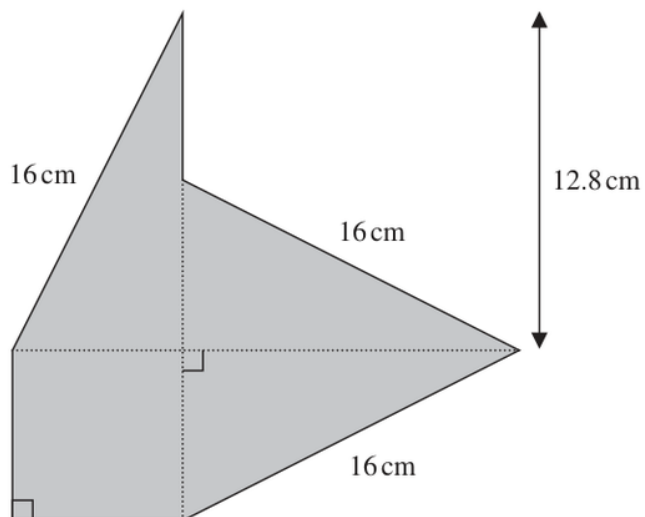
195.

EA

PAST PAPER QUESTION**Q#: 3****Marks: 4****TOPIC: Right-Angled Triangles - Pythagoras & Trigonometry**

Jan 2022 P1HR - Jan 2022 | P1HR

- 3 The shaded shape is made using three identical right-angled triangles and a square.

Diagram **NOT** accurately drawn

Work out the perimeter of the shaded shape.

..... cm

(Total for Question 3 is 4 marks)

PAST PAPER SOLUTION**Q#: 3****Marks: 4****TOPIC: Right-Angled Triangles - Pythagoras & Trigonometry**

Jan 2022 P1HR - Jan 2022 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. Use trigonometry**

Each triangle has hypotenuse 16 cm and one side 12.8 cm.

2. Find the other side

Find the other side:

$$\sqrt{16^2 - 12.8^2} = 9.6 \text{ cm}$$

3. The exposed short vertical side

The exposed short vertical side is

$$12.8 - 9.6 = 3.2 \text{ cm}$$

4. Perimeter

Perimeter:

$$3(16) + 9.6 + 9.6 + 3.2 = 70.4$$

FINAL ANSWER

70.4 cm.

PAST PAPER QUESTION**Q#: 4** **Marks: 4****TOPIC: Graphs of Functions**

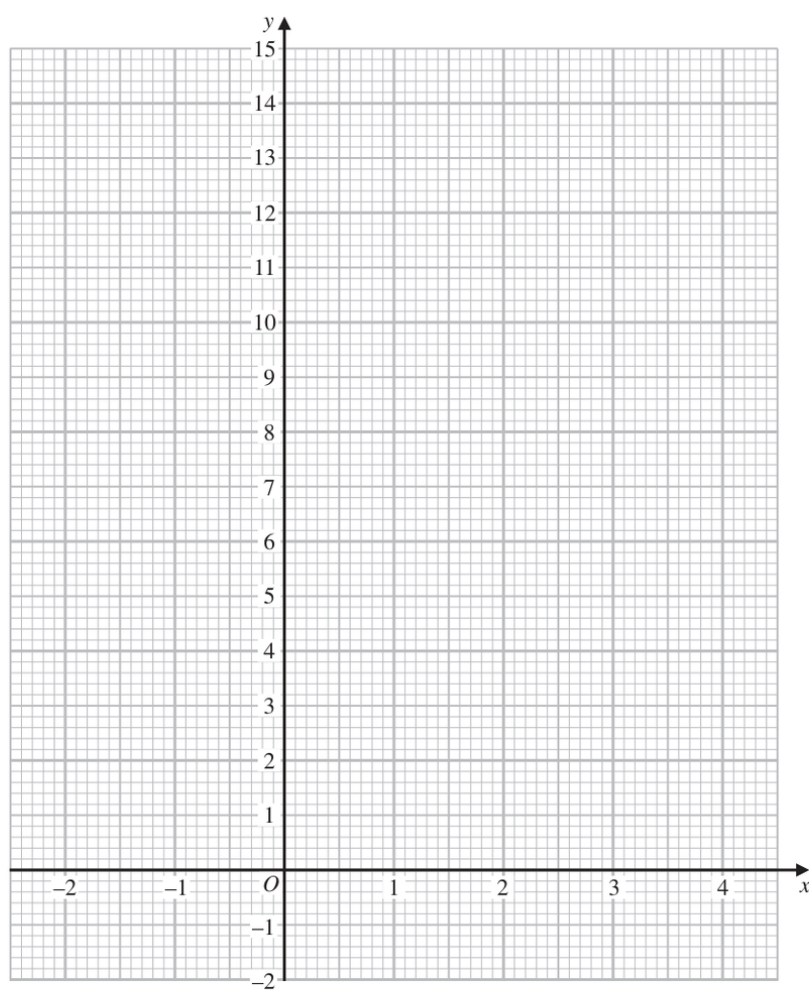
Jan 2022 P1HR - Jan 2022 | P1HR

- 4 (a) Complete the table of values for $y = x^2 - 4x + 3$

x	-2	-1	0	1	2	3	4
y		8	3			0	

(2)

- (b) On the grid, draw the graph of $y = x^2 - 4x + 3$ for values of x from -2 to 4



(2)

(Total for Question 4 is 4 marks)

PAST PAPER SOLUTION

Q#: 4 Marks: 4

TOPIC: **Graphs of Functions**

Jan 2022 P1HR - Jan 2022 | P1HR

WORKED SOLUTION

Dr. Eslam Ahmed

1. Solve quadratic equation

$$y = x^2 - 4x + 3$$

2. The completed table is

The completed table is:

x	-2	-1	0	1	2	3	4
y	15	8	3	0	-1	0	3

3. Plot the points and join

Plot the points and join them with a smooth U-shaped curve.

FINAL ANSWER

Missing values are 15, 0, -1, 3.

PAST PAPER QUESTION**Q#: 5****Marks: 4****TOPIC: Statistics Toolkit**

Jan 2022 P1HR - Jan 2022 | P1HR

5 Yusuf sat 8 examinations.

Here are his marks for 5 of the examinations.

68 72 75 77 80

For his results in all 8 examinations

the mode of his marks is 80

the median of his marks is 74

the range of his marks is 16

Find Yusuf's marks for each of the other 3 examinations.

(Total for Question 5 is 4 marks)

EA

PAST PAPER SOLUTION**Q#: 5****Marks: 4****TOPIC: Statistics Toolkit**

Jan 2022 P1HR - Jan 2022 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. The range is . Since**

The range is 16. Since the greatest known mark is 80,

$$80 - 16 = 64$$

2. Use Statistics Toolkit tag correct

So one missing mark is 64.

3. Calculate statistic

The mode is 80, so another missing mark is 80.

4. Use Statistics Toolkit tag correct

Now the marks in order are

$$64, 68, 72, x, 75, 77, 80, 80$$

5. Calculate statistic

The median is 74, so

$$\frac{x + 75}{2} = 74$$

$$x = 73$$

FINAL ANSWER

64, 73, 80.

PAST PAPER QUESTION**Q#: 6** **Marks: 4****TOPIC: Prime Factors, HCF & LCM**

Jan 2022 P1HR - Jan 2022 | P1HR

- 6** (a) Work out the lowest common multiple (LCM) of 36 and 120

.....
(2)

$$A = 5^2 \times 7^4 \times 11^p$$

$$B = 5^m \times 7^{n-5} \times 11$$

m , n and p are integers such that

$$m > 2$$

$$n > 10$$

$$p > 1$$

- (b) Find the highest common factor (HCF) of A and B
Give your answer as a product of powers of its prime factors.

.....
(2)

.....
(Total for Question 6 is 4 marks)

PAST PAPER SOLUTION**Q#: 6****Marks: 4****TOPIC: Prime Factors, HCF & LCM**

Jan 2022 P1HR - Jan 2022 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. Find the LCM**

$$36 = 2^2 \times 3^2$$

$$120 = 2^3 \times 3 \times 5$$

2. Find the LCM

For the LCM, take the highest power of each prime:

$$\text{LCM} = 2^3 \times 3^2 \times 5 = 360$$

3. Find the LCM

For part (b),

$$A = 5^2 \times 7^4 \times 11^p$$

$$B = 5^m \times 7^{n-5} \times 11$$

4. Find the LCM

Since $m > 2$, the smaller power of 5 is 5^2 .

5. Find the LCM

Since $n > 10$, $n - 5 > 5$, so the smaller power of 7 is 7^4 .

6. Find the LCM

Since $p > 1$, the smaller power of 11 is 11^1 .

$$\text{HCF} = 5^2 \times 7^4 \times 11$$

FINAL ANSWER

(a) 360, (b) $5^2 \times 7^4 \times 11$

PAST PAPER QUESTION**Q#: 7** **Marks: 4****TOPIC: Standard & Compound Units**

Jan 2022 P1HR - Jan 2022 | P1HR

7 Milly went on a car journey.

She travelled from Anesey to Breigh to Clando and then to Duckbridge.

For Anesey to Breigh, Milly drove the 245 km in 2.5 hours.

For Breigh to Clando, Milly drove the 220 km at an average speed of 80 km/h

For Clando to Duckbridge, Milly drove at an average speed of 72 km/h in 50 minutes.

Work out Milly's average speed, in km/h, for the journey from Anesey to Duckbridge.

Give your answer correct to one decimal place.

..... km/h

(Total for Question 7 is 4 marks)

EA

PAST PAPER SOLUTION**Q#: 7** **Marks: 4****TOPIC: Standard & Compound Units**

Jan 2022 P1HR - Jan 2022 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. Calculate value**

First part:

$$\text{distance} = 245 \text{ km}, \quad \text{time} = 2.5 \text{ h}$$

2. Second part

Second part:

$$\text{time} = \frac{220}{80} = 2.75 \text{ h}$$

3. Third part

Third part:

$$50 \text{ minutes} = \frac{5}{6} \text{ h}$$

$$\text{distance} = 72 \times \frac{5}{6} = 60 \text{ km}$$

4. Total distance

Total distance:

$$245 + 220 + 60 = 525 \text{ km}$$

5. Total time

Total time:

$$2.5 + 2.75 + \frac{5}{6} = 6.0833 \dots \text{ h}$$

6. Average speed

Average speed:

$$\frac{525}{6.0833 \dots} = 86.301 \dots$$

FINAL ANSWER

86.3 km/h.

PAST PAPER QUESTION**Q#: 8****Marks: 4****TOPIC: Powers, Roots & Standard Form**

Jan 2022 P1HR - Jan 2022 | P1HR

8 (a) Write 5×10^4 as an ordinary number.

.....
(1)

(b) Write 0.00006 in standard form.

.....
(1)

(c) Work out $(4 \times 10^{512}) \div (1.6 \times 10^{700})$
Give your answer in standard form.

.....
(2)

(Total for Question 8 is 4 marks)

PAST PAPER SOLUTION**Q#: 8****Marks: 4****TOPIC: Powers, Roots & Standard Form**

Jan 2022 P1HR - Jan 2022 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. Convert standard form**

$$5 \times 10^4 = 50000$$

$$0.00006 = 6 \times 10^{-5}$$

2. Convert standard form

For part (c),

$$\begin{aligned} \frac{4 \times 10^{512}}{1.6 \times 10^{700}} &= \frac{4}{1.6} \times 10^{512-700} \\ &= 2.5 \times 10^{-188} \end{aligned}$$

FINAL ANSWER(a) 50000, (b) 6×10^{-5} , (c) 2.5×10^{-188}

PAST PAPER QUESTION**Q#: 9****Marks: 5****TOPIC: Factorising**

Jan 2022 P1HR - Jan 2022 | P1HR

9 (a) Simplify $x^4 \times x^5$
(1)(b) Simplify $(4y^2)^3$
(2)(c) Factorise $n^2 - 7n + 12$
(2)

(Total for Question 9 is 5 marks)

EA

PAST PAPER SOLUTION**Q#: 9****Marks: 5****TOPIC: Factorising**

Jan 2022 P1HR - Jan 2022 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. (a)**

(a) $x^4 \times x^5 = x^9$.

2. (b)

(b) $(4y^2)^3 = 64y^6$.

3. (c)

(c)

$$n^2 - 7n + 12 = (n - 3)(n - 4)$$

FINAL ANSWER(a) x^9 , (b) $64y^6$, (c) $(n - 3)(n - 4)$

PAST PAPER QUESTION**Q#: 10** **Marks: 6****TOPIC: Volume & Surface Area**

Jan 2022 P1HR - Jan 2022 | P1HR

10 Jonty has a storage container in the shape of a cuboid, as shown in the diagram.

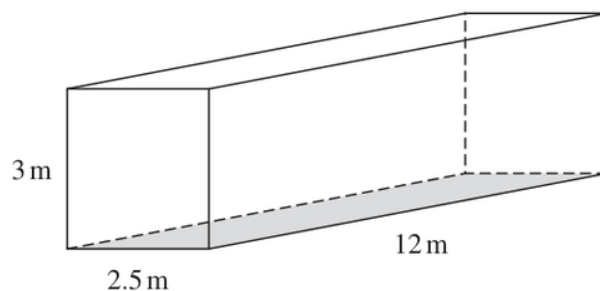


Diagram **NOT**
accurately drawn

Jonty is going to paint the outside of his storage container, apart from the base which is shown shaded in the diagram.

He needs enough paint to cover the four sides and the top.

Each tin of paint covers an area of 15 m^2

The cost of each tin of paint recently increased by 10%

After the increase, the cost of each tin of paint is £26.95

Jonty says

“**Before** the increase, I could have bought enough paint for less than £200”

Show that Jonty is correct.

Show your working clearly.

(Total for Question 10 is 6 marks)

PAST PAPER SOLUTION**Q#: 10****Marks: 6****TOPIC: Volume & Surface Area**

Jan 2022 P1HR - Jan 2022 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. Jonty paints the four sides**

Jonty paints the four sides and the top, but not the base.

2. Top

Top:

$$12 \times 2.5 = 30$$

3. Two long sides

Two long sides:

$$2(12 \times 3) = 72$$

4. Two short sides

Two short sides:

$$2(2.5 \times 3) = 15$$

5. Total area

Total area:

$$30 + 72 + 15 = 117 \text{ m}^2$$

6. Calculate areaEach tin covers 15 m^2 , so the number of tins needed is

$$\frac{117}{15} = 7.8$$

7. He must buy tins

He must buy 8 tins.

8. Calculate area

After the 10% increase, each tin costs £26.95. Before the increase:

$$\frac{26.95}{1.10} = 24.50$$

9. Calculate area

Before the increase, 8 tins would cost

$$8 \times 24.50 = 196$$

10. Calculate area

Since $\pounds 196 < \pounds 200$, Jonty is correct.

FINAL ANSWER

Jonty is correct; before the increase the paint would have cost $\pounds 196$.

EA

PAST PAPER QUESTION**Q#: 11** **Marks: 2****TOPIC: Circles, Arcs & Sectors**

Jan 2022 P1HR - Jan 2022 | P1HR

- 11** The diagram shows sector OPQ of a circle, centre O

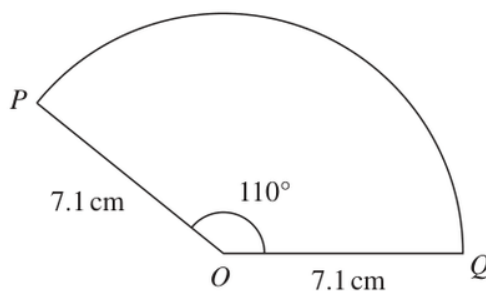


Diagram **NOT**
accurately drawn

$$OP = OQ = 7.1 \text{ cm}$$

$$\text{Angle } POQ = 110^\circ$$

Calculate the area of sector OPQ

Give your answer correct to one decimal place.

..... cm^2

(Total for Question 11 is 2 marks)

PAST PAPER SOLUTION**Q#: 11****Marks: 2****TOPIC: Circles, Arcs & Sectors**

Jan 2022 P1HR - Jan 2022 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. Calculate area**

$$\begin{aligned}\text{area} &= \frac{110}{360} \times \pi(7.1)^2 \\ &= 48.390 \dots\end{aligned}$$

FINAL ANSWER48.4 cm².

EA

PAST PAPER QUESTION**Q#: 12****Marks: 5****TOPIC: Algebraic Fractions**

Jan 2022 P1HR - Jan 2022 | P1HR

12 (a) Expand and simplify $n(n - 4)(3n + 5)$
(2)

(b) Express

$$\frac{3}{x} + \frac{x+2}{2x} + \frac{1}{4}$$

as a single fraction in its simplest form.

.....
(3)

(Total for Question 12 is 5 marks)

EA

PAST PAPER SOLUTION**Q#: 12** **Marks: 5**TOPIC: **Algebraic Fractions**

Jan 2022 P1HR - Jan 2022 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. (a)**

(a)

$$n(n-4)(3n+5) = n(3n^2 - 7n - 20) = 3n^3 - 7n^2 - 20n$$

2. (b)

(b)

$$\frac{3}{x} + \frac{x+2}{2x} + \frac{1}{4}$$

3. Use denominatorUse denominator $4x$:

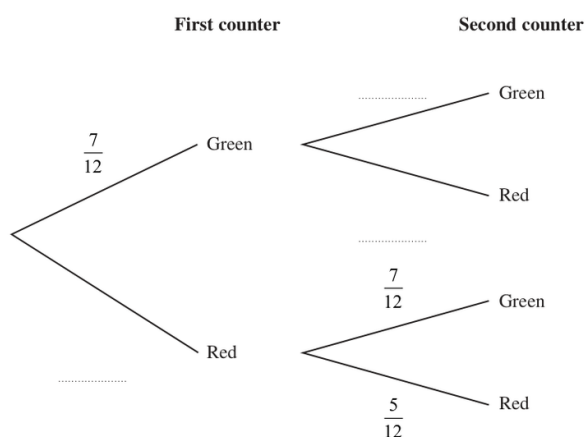
$$\begin{aligned} \frac{12}{4x} + \frac{2(x+2)}{4x} + \frac{x}{4x} &= \frac{12 + 2x + 4 + x}{4x} \\ &= \frac{3x + 16}{4x} \end{aligned}$$

FINAL ANSWER(a) $3n^3 - 7n^2 - 20n$, (b) $\frac{3x+16}{4x}$

PAST PAPER QUESTION**Q#: 13** **Marks: 7****TOPIC: Probability Diagrams - Venn & Tree Diagrams**

Jan 2022 P1HR - Jan 2022 | P1HR

- 13 Hector has a bag that contains 12 counters.
There are 7 green counters and 5 red counters in the bag.
- Hector takes at random a counter from the bag.
He looks at the counter and puts the counter back into the bag.
- Hector then takes at random a second counter from the bag.
He looks at the counter and puts the counter back into the bag.
- (a) Complete the probability tree diagram.



(2)

- (b) Work out the probability that both counters are red.

(2)

Meghan has a jar containing 15 counters.
There are only blue counters, green counters and red counters in the jar.

Hector is going to take at random one of the counters from his bag of 12 counters.
He will look at the counter and put the counter back into the bag.

Hector is then going to take at random a second counter from his bag.
He will look at the counter and put the counter back into the bag.

Meghan is then going to take at random one of the counters from her jar of counters.
She will look at the counter and put the counter back into the jar.

The probability that the 3 counters each have a different colour is $\frac{7}{24}$

- (c) Work out how many blue counters there are in the jar.

(3)

(Total for Question 13 is 7 marks)

PAST PAPER SOLUTION**Q#: 13** **Marks: 7****TOPIC: Probability Diagrams - Venn & Tree Diagrams**

Jan 2022 P1HR - Jan 2022 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. Hector's probability of taking**

Hector's probability of taking one green and one red is

$$2 \times \frac{7}{12} \times \frac{5}{12} = \frac{35}{72}$$

2. Calculate probability

For all three counters to be different, Meghan must take a blue counter.

3. Let the number of blueLet the number of blue counters in Meghan's jar be b . There are 15 counters in the jar.

$$\frac{35}{72} \times \frac{b}{15} = \frac{7}{24}$$

$$\frac{b}{15} = \frac{3}{5}$$

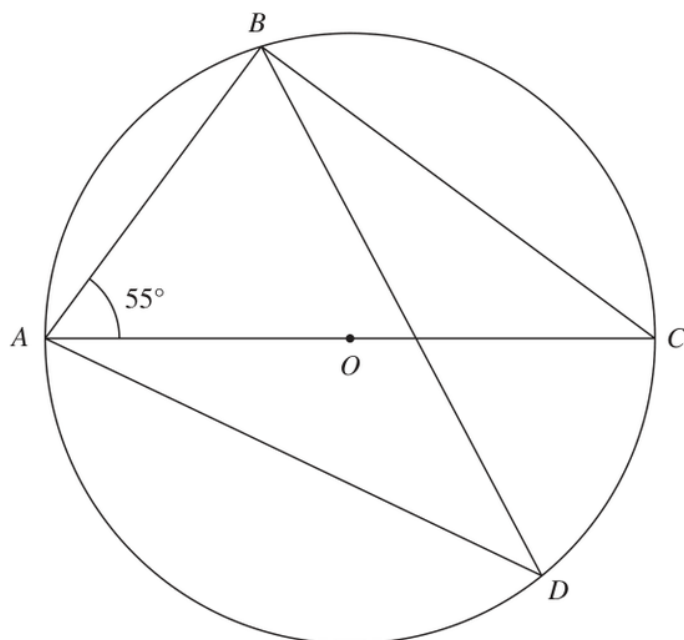
$$b = 9$$

FINAL ANSWER

9 blue counters.

PAST PAPER QUESTION**Q#: 14** **Marks: 4****TOPIC: Circle Theorems**

Jan 2022 P1HR - Jan 2022 | P1HR

14Diagram **NOT**
accurately drawn

A , B , C and D are points on a circle, centre O
 AOC is a diameter of the circle.

Angle $BAC = 55^\circ$

Work out the size of angle ADB

Give a reason for each stage of your working.

o

(Total for Question 14 is 4 marks)

PAST PAPER SOLUTION**Q#: 14** **Marks: 4**TOPIC: **Circle Theorems**

Jan 2022 P1HR - Jan 2022 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. Use trigonometry**Since AC is a diameter,

$$\angle ABC = 90^\circ$$

2. Use trigonometryIn triangle ABC ,

$$\angle ACB = 180^\circ - 90^\circ - 55^\circ = 35^\circ$$

3. Angles in the same segment

Angles in the same segment are equal, so

$$\angle ADB = \angle ACB = 35^\circ$$

FINAL ANSWER 35° .

PAST PAPER QUESTION

Q#: 15 Marks: 3

TOPIC: **Algebraic Proof**

Jan 2022 P1HR - Jan 2022 | P1HR

- 15** Using algebra, prove that, given any 3 consecutive whole numbers, the sum of the square of the smallest number and the square of the largest number is always 2 more than twice the square of the middle number.

(Total for Question 15 is 3 marks)

EA

PAST PAPER SOLUTION**Q#: 15****Marks: 3****TOPIC: Algebraic Proof**

Jan 2022 P1HR - Jan 2022 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. Let the three consecutive whole**

Let the three consecutive whole numbers be

$$n, \quad n + 1, \quad n + 2$$

2. The sum of the squares

The sum of the squares of the smallest and largest is

$$n^2 + (n + 2)^2 = n^2 + n^2 + 4n + 4 = 2n^2 + 4n + 4$$

3. Twice the square of the

Twice the square of the middle number is

$$2(n + 1)^2 = 2(n^2 + 2n + 1) = 2n^2 + 4n + 2$$

4. Calculate value

So

$$2n^2 + 4n + 4 = (2n^2 + 4n + 2) + 2$$

FINAL ANSWER

The sum is always 2 more than twice the square of the middle number.

PAST PAPER QUESTION

Q#: 16

Marks: 4

TOPIC: Sequences

Jan 2022 P1HR - Jan 2022 | P1HR

16 An arithmetic series has first term 1 and common difference 4

Find the sum of all terms of the series from the 41st term to the 100th term inclusive.

(Total for Question 16 is 4 marks)

EA

PAST PAPER SOLUTION**Q#: 16** **Marks: 4**TOPIC: **Sequences**

Jan 2022 P1HR - Jan 2022 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. Find the nth term**

The first term is 1 and the common difference is 4.

2. The 41st term is

The 41st term is

$$1 + 40(4) = 161$$

3. The 100th term is

The 100th term is

$$1 + 99(4) = 397$$

4. There are

There are

$$100 - 41 + 1 = 60$$

5. Find the nth term

terms from the 41st to the 100th inclusive.

$$S = \frac{60}{2}(161 + 397)$$

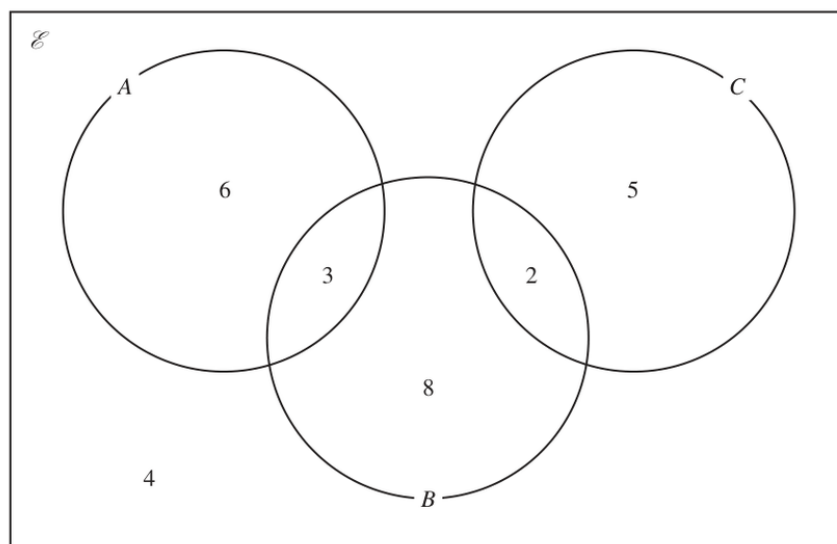
$$S = 30 \times 558 = 16740$$

FINAL ANSWER

16740.

PAST PAPER QUESTION**Q#: 17** **Marks: 4****TOPIC: Set Notation & Venn Diagrams**

Jan 2022 P1HR - Jan 2022 | P1HR

17 The Venn diagram shows a universal set $\mathcal{E}$ and three sets A , B and C .6, 3, 8, 2, 5 and 4 represent the **numbers** of elements.

Find

(i) $n(A \cup B)$

.....
(1)

(ii) $n(A \cap C)$

.....
(1)

(iii) $n(B \cap C')$

.....
(1)

(iv) $n(A' \cup B' \cup C')$

.....
(1)**(Total for Question 17 is 4 marks)**

PAST PAPER SOLUTION**Q#: 17****Marks: 4****TOPIC: Set Notation & Venn Diagrams**

Jan 2022 P1HR - Jan 2022 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. From the Venn diagram**

From the Venn diagram:

$$A \text{ only} = 6, \quad A \cap B = 3, \quad B \text{ only} = 8, \quad B \cap C = 2, \quad C \text{ only} = 5, \quad \text{outside} = 4$$

2. Calculate valueThere is no region in both A and C , and there is no triple intersection.

$$n(A \cup B) = 6 + 3 + 8 + 2 = 19$$

$$n(A \cap C) = 0$$

$$n(B \cap C') = 3 + 8 = 11$$

3. UseFor $A' \cup B' \cup C'$, use

$$A' \cup B' \cup C' = (A \cap B \cap C)'$$

4. SetSince $A \cap B \cap C$ is empty, this includes everyone in the universal set.

$$6 + 3 + 8 + 2 + 5 + 4 = 28$$

FINAL ANSWER

(i) 19, (ii) 0, (iii) 11, (iv) 28

PAST PAPER QUESTION

Q#: 18

Marks: 4

TOPIC: **Area & Volume of Similar Shapes**

Jan 2022 P1HR - Jan 2022 | P1HR

18 The three solids **A**, **B** and **C** are similar such that

the surface area of **A** : the surface area of **B** = 4 : 9

and

the volume of **B** : the volume of **C** = 125 : 343

Work out the ratio

the height of **A** : the height of **C**

Give your ratio in its simplest form.

(Total for Question 18 is 4 marks)

EA

PAST PAPER SOLUTION**Q#: 18****Marks: 4****TOPIC:** Area & Volume of Similar Shapes

Jan 2022 P1HR - Jan 2022 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. Split the ratio**Surface area ratio $A : B = 4 : 9$, so the length ratio is

$$\sqrt{4} : \sqrt{9} = 2 : 3$$

2. Split the ratioVolume ratio $B : C = 125 : 343$, so the length ratio is

$$\sqrt[3]{125} : \sqrt[3]{343} = 5 : 7$$

3. Split the ratio

Therefore

$$\frac{A}{C} = \frac{2}{3} \times \frac{5}{7} = \frac{10}{21}$$

FINAL ANSWER

10 : 21

PAST PAPER QUESTION**Q#: 19** **Marks: 3****TOPIC: Completing the Square**

Jan 2022 P1HR - Jan 2022 | P1HR

19 Given that $\left(\sqrt[3]{\frac{1}{x}}\right)^4 = x^m$

(a) find the value of m

$$m = \dots\dots\dots (1)$$

Given that a , b and c are integers,

(b) express $3x^2 + 12x + 19$ in the form $a(x + b)^2 + c$

$$\dots\dots\dots (2)$$

(Total for Question 19 is 3 marks)

PAST PAPER SOLUTION**Q#: 19** **Marks: 3****TOPIC: Completing the Square**

Jan 2022 P1HR - Jan 2022 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. (a)**

(a)

$$\sqrt[3]{\frac{1}{x}} = x^{-1/3}$$

2. Evaluate fractionSo $m = -\frac{1}{3}$.**3. (b)**

(b)

$$3x^2 + 12x + 19 = 3(x^2 + 4x) + 19$$

$$= 3((x + 2)^2 - 4) + 19$$

$$= 3(x + 2)^2 + 7$$

FINAL ANSWER $m = -\frac{1}{3}$, and $3(x + 2)^2 + 7$

PAST PAPER QUESTION**Q#: 20****Marks: 6****TOPIC: Transformations of Graphs**

Jan 2022 P1HR - Jan 2022 | P1HR

20 The curve with equation $y = f(x)$ has one turning point.

The coordinates of this turning point are $(-6, -4)$

(a) Write down the coordinates of the turning point on the curve with equation

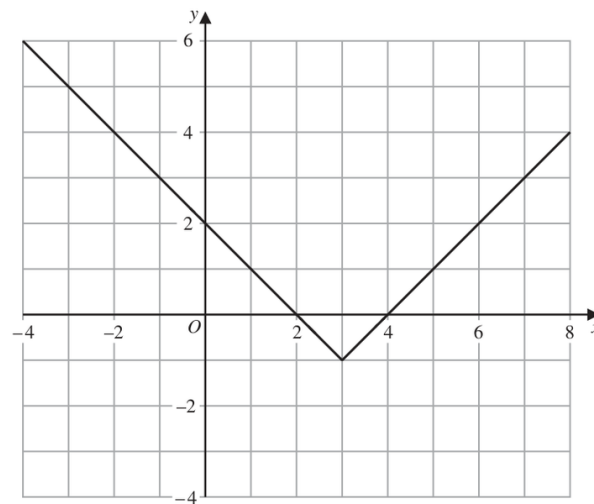
(i) $y = f(x) + 5$

(.....,)

(ii) $y = f(3x)$

(.....,)
(2)

The graph of $y = g(x)$ is shown on the grid below.



(b) On the grid, sketch the graph of $y = 2g(x)$ for $-1 \leq x \leq 7$

(2)

The graph of $y = h(x)$ intersects the x -axis at two points.
The coordinates of the two points are $(-1, 0)$ and $(6, 0)$

The graph of $y = h(x + a)$ passes through the point with coordinates $(2, 0)$, where a is a constant.

(c) Find the two possible values of a

.....,
(2)

(Total for Question 20 is 6 marks)

PAST PAPER SOLUTION**Q#: 20****Marks: 6****TOPIC: Transformations of Graphs**

Jan 2022 P1HR - Jan 2022 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. The turning point of is**The turning point of $y = f(x)$ is $(-6, -4)$.**2. Calculate value**

For

$$y = f(x) + 5$$

3. the graph moves 5 units

the graph moves 5 units up:

$$(-6, -4) \rightarrow (-6, 1)$$

4. Calculate value

For

$$y = f(3x)$$

5. the coordinate is divided bythe x -coordinate is divided by 3:

$$(-6, -4) \rightarrow (-2, -4)$$

6. Read the graphFor the sketch of $y = 2g(x)$, keep every x -coordinate the same and double every y -coordinate.**7. Useful points are**

Useful points are

$$(-1, 3) \rightarrow (-1, 6)$$

$$(2, 0) \rightarrow (2, 0)$$

$$(3, -1) \rightarrow (3, -2)$$

$$(4, 0) \rightarrow (4, 0)$$

$$(7, 3) \rightarrow (7, 6)$$

8. Solve inequality

Draw straight line segments through these points for $-1 \leq x \leq 7$.

9. Calculate value

For part (c), $h(x)$ has roots at $x = -1$ and $x = 6$.

10. Calculate value

For $h(x + a)$ to pass through $(2, 0)$,

$$2 + a = -1 \quad \text{or} \quad 2 + a = 6$$

$$a = -3 \quad \text{or} \quad a = 4$$

FINAL ANSWER

$(-6, 1)$, $(-2, -4)$, sketch through the doubled points, and $a = -3$ or $a = 4$

EA

PAST PAPER QUESTION**Q#: 20** **Marks: 6****TOPIC: Transformations of Graphs**

Jan 2022 P1HR - Jan 2022 | P1HR

20 The curve with equation $y = f(x)$ has one turning point.

The coordinates of this turning point are $(-6, -4)$

(a) Write down the coordinates of the turning point on the curve with equation

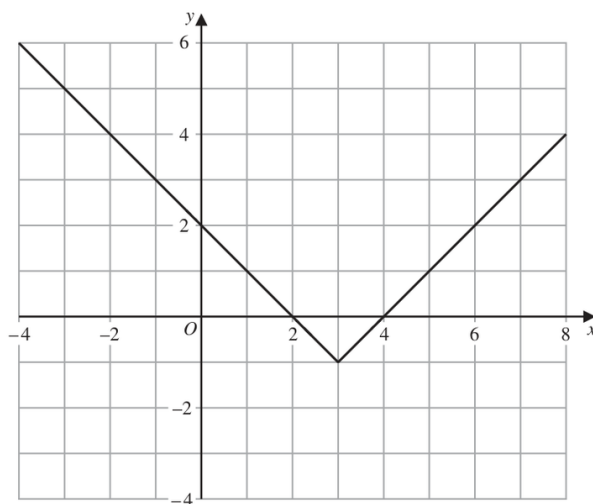
(i) $y = f(x) + 5$

(.....,.....)

(ii) $y = f(3x)$

(.....,.....)
(2)

The graph of $y = g(x)$ is shown on the grid below.



(b) On the grid, sketch the graph of $y = 2g(x)$ for $-1 \leq x \leq 7$

(2)

The graph of $y = h(x)$ intersects the x -axis at two points.
The coordinates of the two points are $(-1, 0)$ and $(6, 0)$

The graph of $y = h(x + a)$ passes through the point with coordinates $(2, 0)$, where a is a constant.

(c) Find the two possible values of a

(2)

(Total for Question 20 is 6 marks)

PAST PAPER SOLUTION**Q#: 20****Marks: 6****TOPIC: Transformations of Graphs**

Jan 2022 P1HR - Jan 2022 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. The turning point of is**The turning point of $y = f(x)$ is $(-6, -4)$.**2. Calculate value**

For

$$y = f(x) + 5$$

3. the graph moves 5 units

the graph moves 5 units up:

$$(-6, -4) \rightarrow (-6, 1)$$

4. Calculate value

For

$$y = f(3x)$$

5. the coordinate is divided bythe x -coordinate is divided by 3:

$$(-6, -4) \rightarrow (-2, -4)$$

6. Read the graphFor the sketch of $y = 2g(x)$, keep every x -coordinate the same and double every y -coordinate.**7. Useful points are**

Useful points are

$$(-1, 3) \rightarrow (-1, 6)$$

$$(2, 0) \rightarrow (2, 0)$$

$$(3, -1) \rightarrow (3, -2)$$

$$(4, 0) \rightarrow (4, 0)$$

$$(7, 3) \rightarrow (7, 6)$$

8. Solve inequality

Draw straight line segments through these points for $-1 \leq x \leq 7$.

9. Calculate value

For part (c), $h(x)$ has roots at $x = -1$ and $x = 6$.

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For $h(x + a)$ to pass through $(2, 0)$,

$$2 + a = -1 \quad \text{or} \quad 2 + a = 6$$

$$a = -3 \quad \text{or} \quad a = 4$$

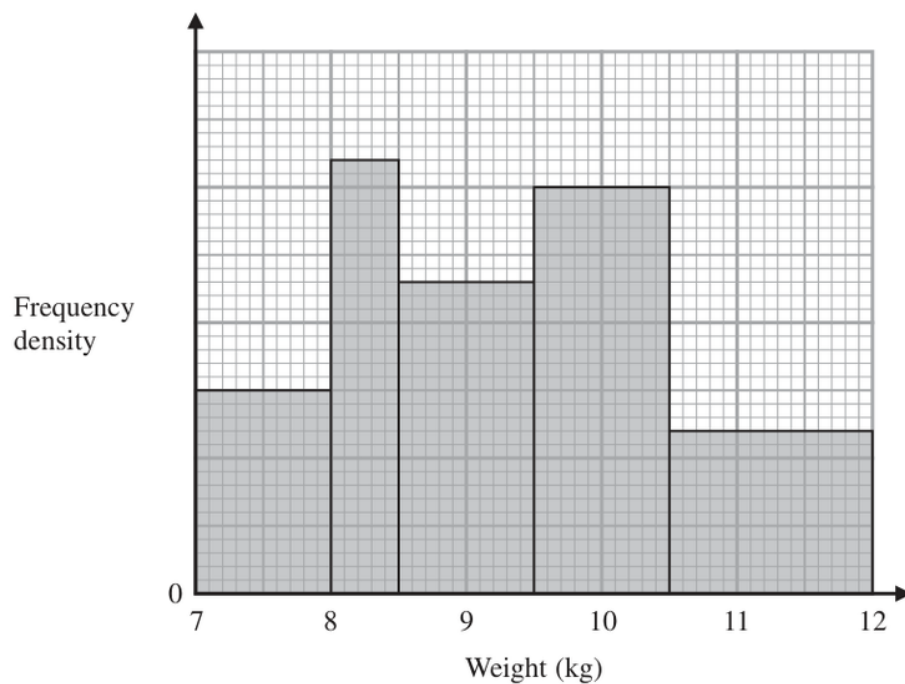
FINAL ANSWER

$(-6, 1)$, $(-2, -4)$, sketch through the doubled points, and $a = -3$ or $a = 4$

EA

PAST PAPER QUESTION**Q#: 21** **Marks: 3****TOPIC: Histograms**

Jan 2022 P1HR - Jan 2022 | P1HR

21

The histogram gives information about the weights, in kg, of all the watermelons in a field.

There are 16 watermelons with a weight between 8 kg and 8.5 kg

Work out the total number of watermelons in the field.

(Total for Question 21 is 3 marks)

PAST PAPER SOLUTION**Q#: 21****Marks: 3****TOPIC: Histograms**

Jan 2022 P1HR - Jan 2022 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. Use histogram**

For a histogram,

$$\text{frequency} = \text{class width} \times \text{frequency density}$$

2. Use frequency tableFor $8 \leq w < 8.5$, the frequency is 16 and the class width is 0.5, so

$$\text{frequency density} = \frac{16}{0.5} = 32$$

3. Use histogram

Using this scale from the histogram, the frequency densities are:

$$15, 32, 23, 30, 12$$

4. The total frequency is

The total frequency is

$$(1)(15) + (0.5)(32) + (1)(23) + (1)(30) + (1.5)(12)$$

$$= 15 + 16 + 23 + 30 + 18$$

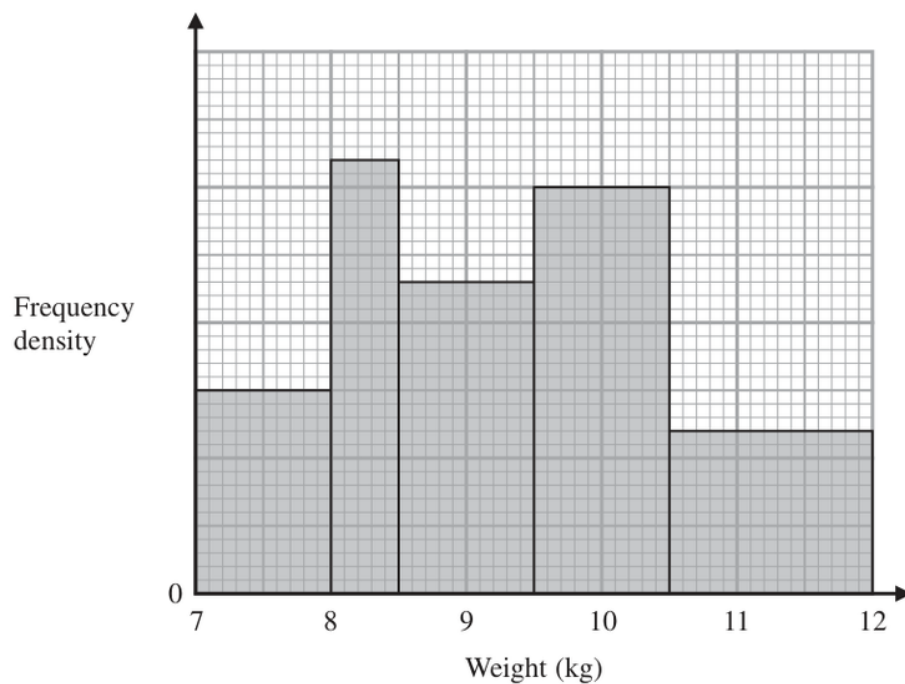
$$= 102$$

FINAL ANSWER

102 watermelons

PAST PAPER QUESTION**Q#: 21** **Marks: 3****TOPIC: Histograms**

Jan 2022 P1HR - Jan 2022 | P1HR

21

The histogram gives information about the weights, in kg, of all the watermelons in a field.

There are 16 watermelons with a weight between 8 kg and 8.5 kg

Work out the total number of watermelons in the field.

(Total for Question 21 is 3 marks)

PAST PAPER SOLUTION**Q#: 21****Marks: 3****TOPIC: Histograms**

Jan 2022 P1HR - Jan 2022 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. Use histogram**

For a histogram,

$$\text{frequency} = \text{class width} \times \text{frequency density}$$

2. Use frequency tableFor $8 \leq w < 8.5$, the frequency is 16 and the class width is 0.5, so

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Using this scale from the histogram, the frequency densities are:

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4. The total frequency is

The total frequency is

$$(1)(15) + (0.5)(32) + (1)(23) + (1)(30) + (1.5)(12)$$

$$= 15 + 16 + 23 + 30 + 18$$

$$= 102$$

FINAL ANSWER

102 watermelons

PAST PAPER QUESTION

Q#: 22

Marks: 4

TOPIC: **Rounding, Estimation & Bounds**

Jan 2022 P1HR - Jan 2022 | P1HR

22 The diagram shows triangle ABC

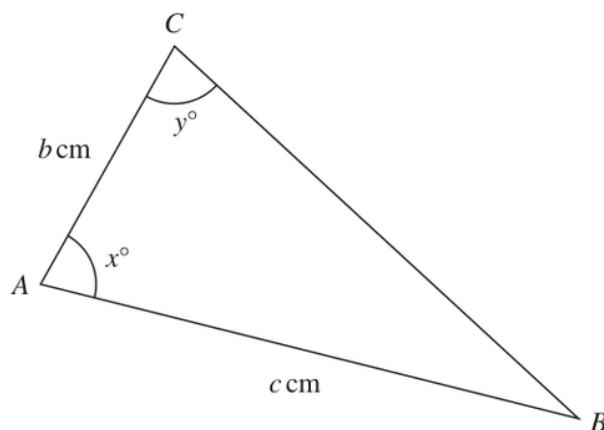


Diagram **NOT**
accurately drawn

$c = 11.5$ correct to one decimal place
 $x = 80$ correct to the nearest whole number
 $y = 75$ correct to the nearest whole number

Calculate the upper bound for the value of b
Show your working clearly.
Give your answer correct to 3 significant figures.

(Total for Question 22 is 4 marks)

PAST PAPER SOLUTION**Q#: 22****Marks: 4****TOPIC: Rounding, Estimation & Bounds**

Jan 2022 P1HR - Jan 2022 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. Find upper bound**For an upper bound for b , use the upper bound of c :

$$c = 11.55$$

2. Use sine ruleUse the lower bounds of x and y , because this makes angle B as large as possible and makes $\sin y$ as small as possible.

$$x = 79.5^\circ, \quad y = 74.5^\circ$$

3. Use trigonometry

So

$$\angle B = 180^\circ - 79.5^\circ - 74.5^\circ = 26^\circ$$

4. Use trigonometry

Using the sine rule,

$$\frac{b}{\sin 26^\circ} = \frac{11.55}{\sin 74.5^\circ}$$

$$b = \frac{11.55 \sin 26^\circ}{\sin 74.5^\circ} = 5.254...$$

FINAL ANSWER

5.25 cm to 3 significant figures.

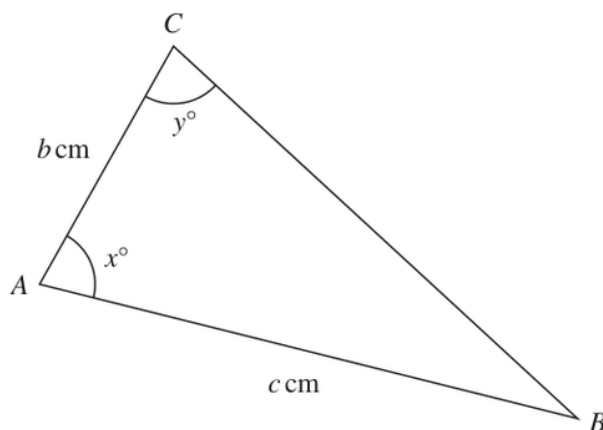
PAST PAPER QUESTION

Q#: 22

Marks: 4

TOPIC: **Rounding, Estimation & Bounds**

Jan 2022 P1HR - Jan 2022 | P1HR

22 The diagram shows triangle ABC Diagram **NOT**
accurately drawn

- $c = 11.5$ correct to one decimal place
 $x = 80$ correct to the nearest whole number
 $y = 75$ correct to the nearest whole number

Calculate the upper bound for the value of b
Show your working clearly.
Give your answer correct to 3 significant figures.

(Total for Question 22 is 4 marks)

PAST PAPER SOLUTION**Q#: 22****Marks: 4****TOPIC: Rounding, Estimation & Bounds**

Jan 2022 P1HR - Jan 2022 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. Find upper bound**For an upper bound for b , use the upper bound of c :

$$c = 11.55$$

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$$x = 79.5^\circ, \quad y = 74.5^\circ$$

3. Use trigonometry

So

$$\angle B = 180^\circ - 79.5^\circ - 74.5^\circ = 26^\circ$$

4. Use trigonometry

Using the sine rule,

$$\frac{b}{\sin 26^\circ} = \frac{11.55}{\sin 74.5^\circ}$$

$$b = \frac{11.55 \sin 26^\circ}{\sin 74.5^\circ} = 5.254...$$

FINAL ANSWER

5.25 cm to 3 significant figures.

PAST PAPER QUESTION**Q#: 23****Marks: 6****TOPIC: Differentiation**

Jan 2022 P1HR - Jan 2022 | P1HR

- 23** Two particles, P and Q , move along a straight line.
The fixed point O lies on this line.

The displacement of P from O at time t seconds is s metres, where

$$s = t^3 - 4t^2 + 5t \quad \text{for } t > 1$$

The displacement of Q from O at time t seconds is x metres, where

$$x = t^2 - 4t + 4 \quad \text{for } t > 1$$

Find the range of values of t where $t > 1$ for which both particles are moving in the same direction along the straight line.

(Total for Question 23 is 6 marks)

PAST PAPER SOLUTION**Q#: 23****Marks: 6****TOPIC: Differentiation**

Jan 2022 P1HR - Jan 2022 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. Particle**Particle P :

$$s = t^3 - 4t^2 + 5t$$

$$\frac{ds}{dt} = 3t^2 - 8t + 5$$

$$\frac{ds}{dt} = (3t - 5)(t - 1)$$

2. ParticleParticle Q :

$$x = t^2 - 4t + 4$$

$$\frac{dx}{dt} = 2t - 4 = 2(t - 2)$$

3. DifferentiateFor $t > 1$:**4. moves in the negative direction**

- P moves in the negative direction when $1 < t < \frac{5}{3}$.
- P moves in the positive direction when $t > \frac{5}{3}$.
- Q moves in the negative direction when $1 < t < 2$.
- Q moves in the positive direction when $t > 2$.

5. Differentiate

So both particles move in the same direction when both velocities have the same sign.

FINAL ANSWER

$$1 < t < \frac{5}{3} \text{ or } t > 2$$

PAST PAPER QUESTION**Q#: 23** **Marks: 6****TOPIC: Differentiation**

Jan 2022 P1HR - Jan 2022 | P1HR

- 23** Two particles, P and Q , move along a straight line.
The fixed point O lies on this line.

The displacement of P from O at time t seconds is s metres, where

$$s = t^3 - 4t^2 + 5t \quad \text{for } t > 1$$

The displacement of Q from O at time t seconds is x metres, where

$$x = t^2 - 4t + 4 \quad \text{for } t > 1$$

Find the range of values of t where $t > 1$ for which both particles are moving in the same direction along the straight line.

(Total for Question 23 is 6 marks)

PAST PAPER SOLUTION**Q#: 23****Marks: 6****TOPIC: Differentiation**

Jan 2022 P1HR - Jan 2022 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. Particle**Particle P :

$$s = t^3 - 4t^2 + 5t$$

$$\frac{ds}{dt} = 3t^2 - 8t + 5$$

$$\frac{ds}{dt} = (3t - 5)(t - 1)$$

2. ParticleParticle Q :

$$x = t^2 - 4t + 4$$

$$\frac{dx}{dt} = 2t - 4 = 2(t - 2)$$

3. DifferentiateFor $t > 1$:**4. moves in the negative direction**

- P moves in the negative direction when $1 < t < \frac{5}{3}$.
- P moves in the positive direction when $t > \frac{5}{3}$.
- Q moves in the negative direction when $1 < t < 2$.
- Q moves in the positive direction when $t > 2$.

5. Differentiate

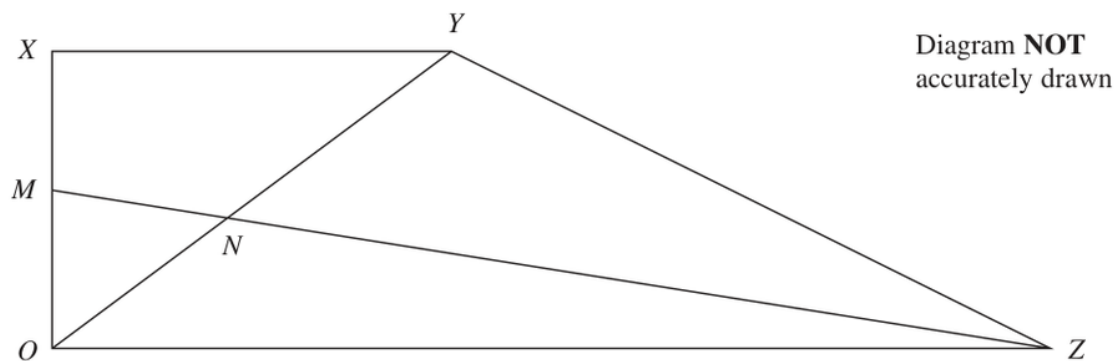
So both particles move in the same direction when both velocities have the same sign.

FINAL ANSWER

$$1 < t < \frac{5}{3} \text{ or } t > 2$$

PAST PAPER QUESTION**Q#: 24** **Marks: 5****TOPIC: Vectors**

Jan 2022 P1HR - Jan 2022 | P1HR

24 $OXYZ$ is a trapezium.

$$\overrightarrow{OX} = \mathbf{a}$$

$$\overrightarrow{XY} = \mathbf{b}$$

$$\overrightarrow{OZ} = 3\mathbf{b}$$

M is the midpoint of OX

N is the point such that MNZ and ONY are straight lines.

Given that $ON : OY = \lambda : 1$

use a vector method to find the value of λ

$$\lambda = \dots\dots\dots$$

(Total for Question 24 is 5 marks)

PAST PAPER SOLUTION**Q#: 24****Marks: 5**TOPIC: **Vectors**

Jan 2022 P1HR - Jan 2022 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. Take as the origin**Take O as the origin.

$$\vec{OX} = \mathbf{a}, \quad \vec{XY} = \mathbf{b}, \quad \vec{OZ} = 3\mathbf{b}$$

2. Use vectors

So

$$\vec{OY} = \mathbf{a} + \mathbf{b}$$

3. Find the midpointSince M is the midpoint of OX ,

$$\vec{OM} = \frac{1}{2}\mathbf{a}$$

4. Use vectorsGiven $ON : OY = \lambda : 1$,

$$\vec{ON} = \lambda(\mathbf{a} + \mathbf{b})$$

5. Use vectorsPoint N also lies on MZ , so

$$\vec{ON} = \frac{1}{2}\mathbf{a} + t\left(3\mathbf{b} - \frac{1}{2}\mathbf{a}\right)$$

$$\vec{ON} = \left(\frac{1}{2} - \frac{t}{2}\right)\mathbf{a} + 3t\mathbf{b}$$

6. Use vectorsEquate coefficients with $\lambda\mathbf{a} + \lambda\mathbf{b}$:

$$\lambda = \frac{1}{2} - \frac{t}{2}$$

$$\lambda = 3t$$

7. Use vectors

So

$$t = \frac{\lambda}{3}$$

8. Substitute

Substitute:

$$\lambda = \frac{1}{2} - \frac{1}{2} \left(\frac{\lambda}{3} \right)$$

$$\lambda = \frac{1}{2} - \frac{\lambda}{6}$$

$$\frac{7}{6}\lambda = \frac{1}{2}$$

$$\lambda = \frac{3}{7}$$

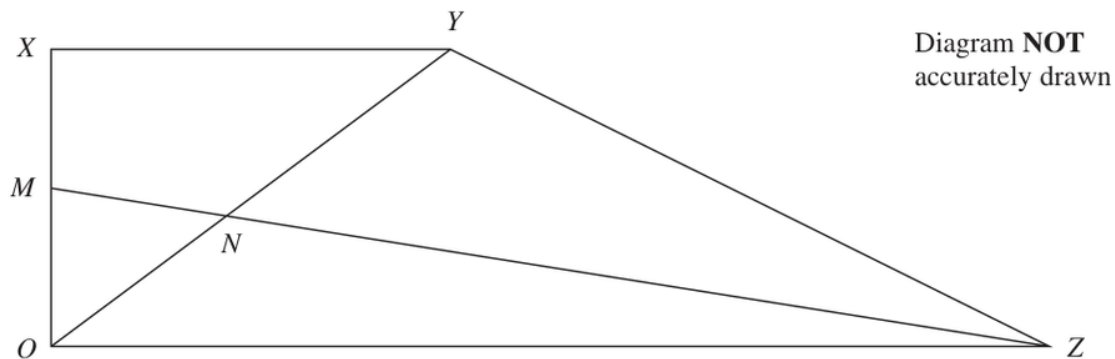
FINAL ANSWER

$$\lambda = \frac{3}{7}$$

EA

PAST PAPER QUESTION**Q#: 24** **Marks: 5****TOPIC: Vectors**

Jan 2022 P1HR - Jan 2022 | P1HR

24 $OXYZ$ is a trapezium.

$$\overrightarrow{OX} = \mathbf{a}$$

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M is the midpoint of OX

N is the point such that MNZ and ONY are straight lines.

Given that $ON : OY = \lambda : 1$

use a vector method to find the value of λ

$$\lambda = \dots\dots\dots$$

(Total for Question 24 is 5 marks)

PAST PAPER SOLUTION

Q#: 24

Marks: 5

TOPIC: **Vectors**

Jan 2022 P1HR - Jan 2022 | P1HR

WORKED SOLUTION

Dr. Eslam Ahmed

1. Take as the originTake O as the origin.

$$\vec{OX} = \mathbf{a}, \quad \vec{XY} = \mathbf{b}, \quad \vec{OZ} = 3\mathbf{b}$$

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So

$$\vec{OY} = \mathbf{a} + \mathbf{b}$$

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5. Use vectorsPoint N also lies on MZ , so

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$$\vec{ON} = \left(\frac{1}{2} - \frac{t}{2}\right)\mathbf{a} + 3t\mathbf{b}$$

6. Use vectorsEquate coefficients with $\lambda\mathbf{a} + \lambda\mathbf{b}$:

$$\lambda = \frac{1}{2} - \frac{t}{2}$$

$$\lambda = 3t$$

7. Use vectors

So

$$t = \frac{\lambda}{3}$$

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Substitute:

$$\lambda = \frac{1}{2} - \frac{1}{2} \left(\frac{\lambda}{3} \right)$$

$$\lambda = \frac{1}{2} - \frac{\lambda}{6}$$

$$\frac{7}{6}\lambda = \frac{1}{2}$$

$$\lambda = \frac{3}{7}$$

FINAL ANSWER

$$\lambda = \frac{3}{7}$$

EA